CE586 EARTHQUAKE ENGINEERING
2025-2026 SPRING SEMESTER

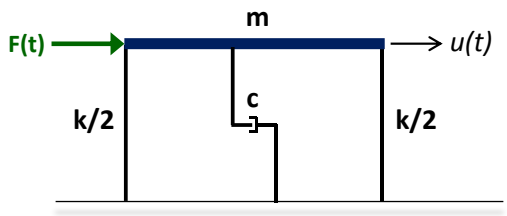
ASSIGNMENT #2 (DUE 14 APRIL)

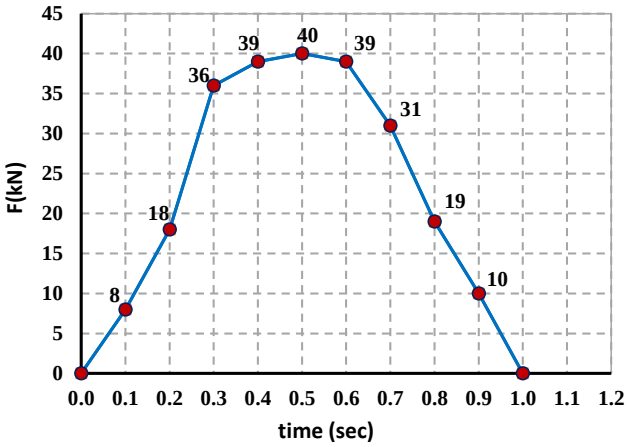
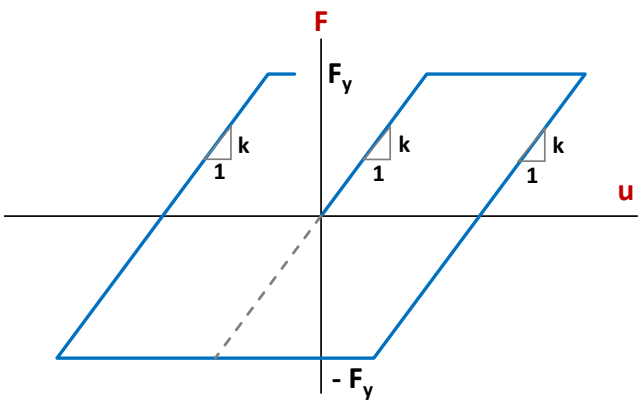
Q1. A linear-elastic SDOF system with m=3600 kg, k=1400 kN/m and c=70 kN.s/m is subjected to the dynamic loading provided in the below plot. The system is initially at rest.

a) Compute the dynamic response of the system for $t \leq 2.0$ seconds by using the *Newmark-Beta Numerical Method* with constant average acceleration assumption. Take time step $\Delta t = 0.1$ seconds and obtain the displacement, velocity and acceleration values at each time step. Plot the graphics of displacement, velocity and acceleration w.r.t. time.

b) Is there a change in the response if $\Delta t = 0.01$ seconds? Plot curves with both time steps on the same plot, compare and discuss briefly.

You can use any programming language, MatLab or even MS Excel. Please submit your code together with the solution.





Q2. If the system in the previous question has an elasto-plastic force-displacement relationship as shown below with F_y=24 kN, solve the same problem for the inelastic case and plot displacement, velocity and acceleration time series (for $t \leq 2.0$ seconds and $\Delta t = 0.1$ seconds) and compare these by the dynamic solutions for linear-elastic case of Question 3. Also sketch the force-displacement graphs of linear-elastic case (Question 3) and the elasto-plastic case (this question) on the same plot. Briefly discuss all the results obtained.